

Designing Optimal Secure Hardware IPs for Consumer Electronics using Fingerprint Watermark and Evolutionary Algorithm Framework

Presented in IEEE International Conference on Future Technology for Smart Society (ICFTSS)

Anirban Sengupta, Vsihal Chourasia, Nilesh Kumar and Praveen Dawar,
"Designing Optimal Secure Hardware IPs for Consumer Electronics using Fingerprint Watermark and Evolutionary Algorithm Framework," 2024 ICFTSS, Kuala Lumpur, Malaysia, pp. 123-127, 2024

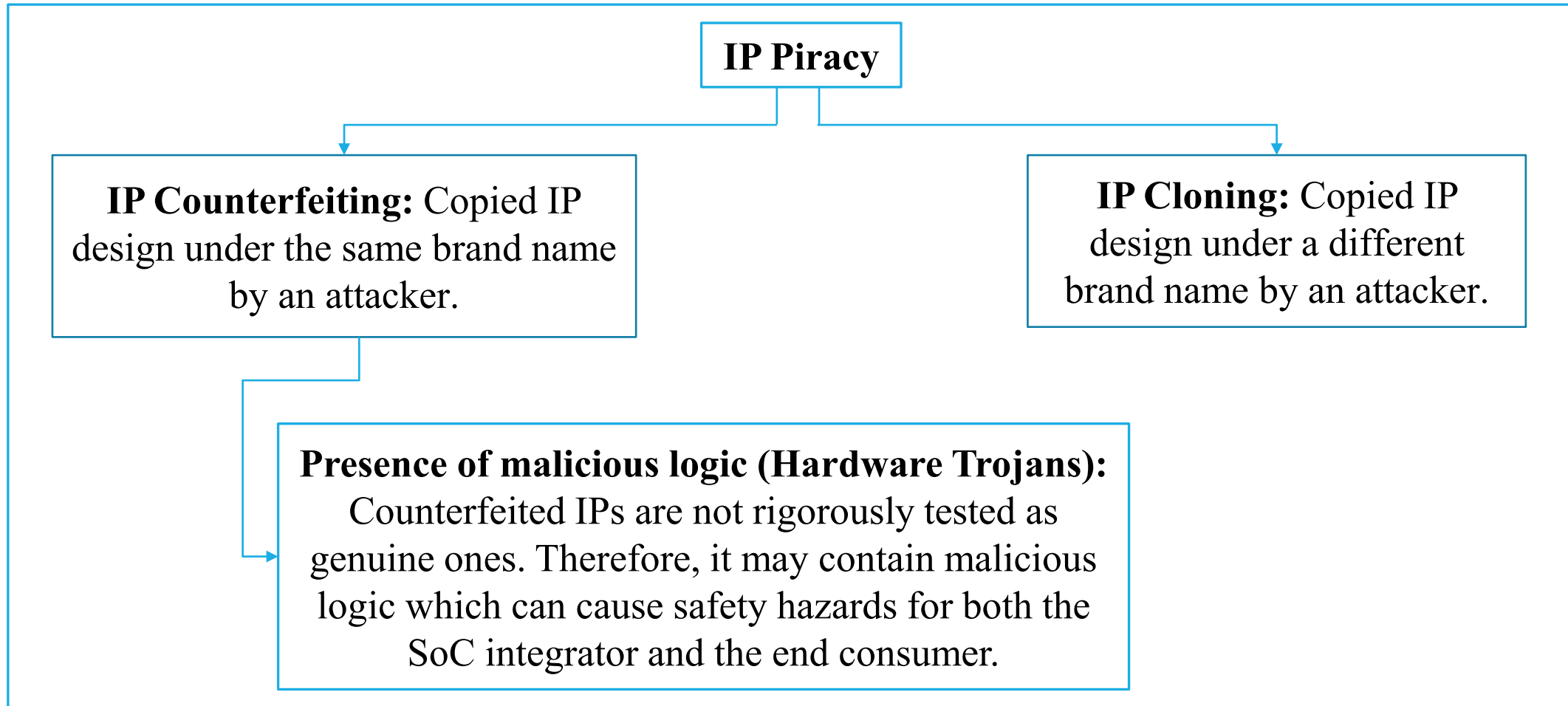
INTRODUCTION

- The digital world is being revolutionized by the advent of smart cities.
- Cybersecurity in smart cities, smart homes, and smart societies is essential for ensuring the safety and reliability of end users.
- Holistic cybersecurity includes [1][2]:
 - Countermeasures against malicious hardware
 - Protection against malicious, untrustworthy, or pirated hardware



Fig. 1 : Hardware (IC) design process

HARDWARE THREATS



PREVIOUS WORKS

Sr. No.	Existing Work	Technique Used	Remarks
1.	Sengupta et. al., [4] (2016)	Multi-variable signature encoding scheme for embedding a dynamic watermark	[4] provides weaker security than the proposed approach.
2.	Sengupta et. al., [5] (2021)	Secure CNN-hardware accelerator design flow using IP seller's facial biometric	[5] presents a design flow for a dedicated hardware accelerator for convolutional layer IP. However, it depicts weaker security strength compared to the proposed approach due to the generation of fewer security constraints.
3.	F. Koushanfar et al. [6] (2005)	synthesis directives to create watermarks	[6] does not provide any security analysis in terms of known hardware threads. Moreover [6] does not provide any comparative analysis against state of the art IP watermarking techniques, unlike the proposed approach.

PROPOSED WORK

- The proposed approach introduces a novel HLS based low cost security algorithm.
- It presents a biometric fingerprint-based watermarking methodology for securing IP design.
- The proposed approach exploits genetic algorithm based design space exploration to explore low-cost resource configuration along with optimal fingerprint minutiae point set and unroll factor.
- This approach provides a robust countermeasure against piracy and counterfeiting.

PROPOSED WORK : Flow Diagram

Flow diagram of the proposed approach

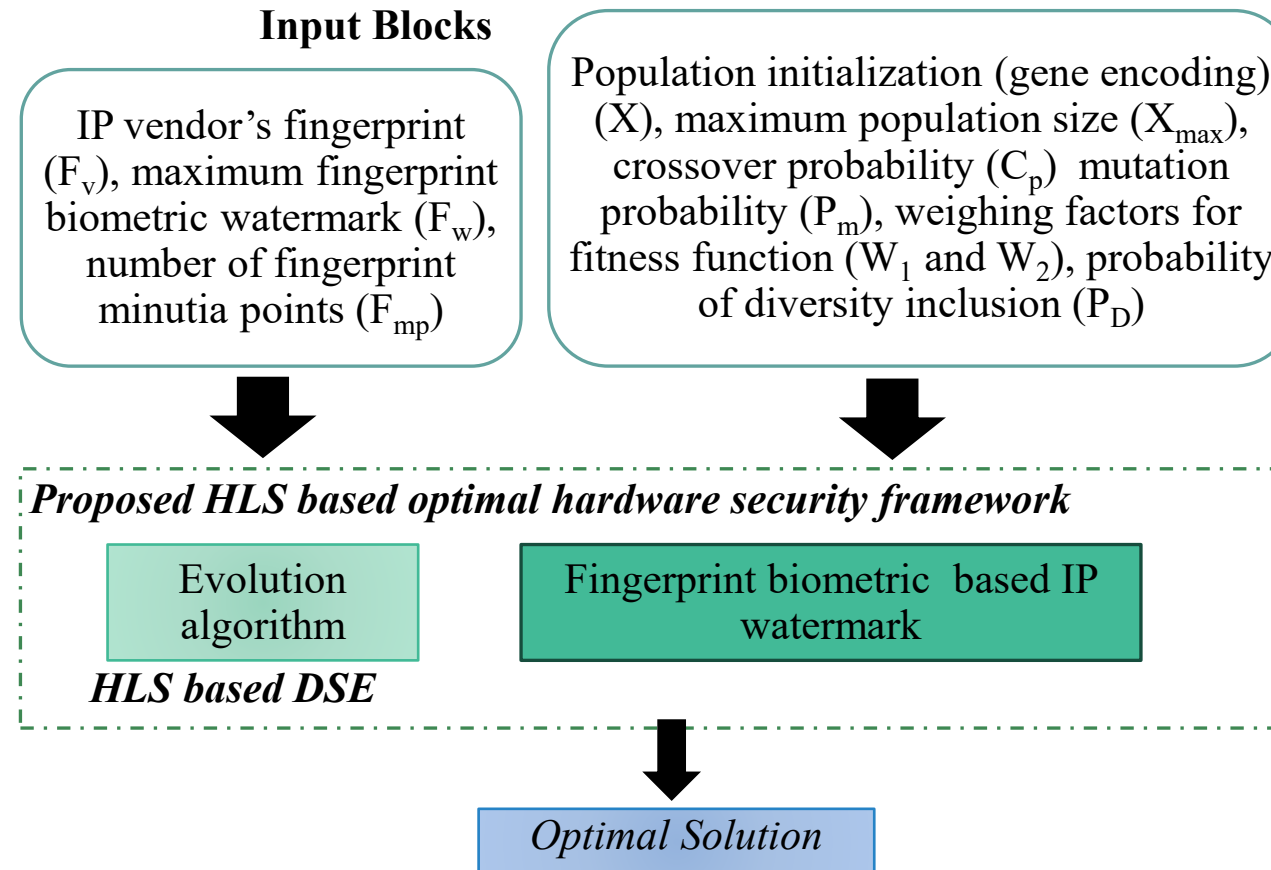


Fig. 2. Overview of proposed approach

PROPOSED WORK CONTINUE...

Design space exploration:

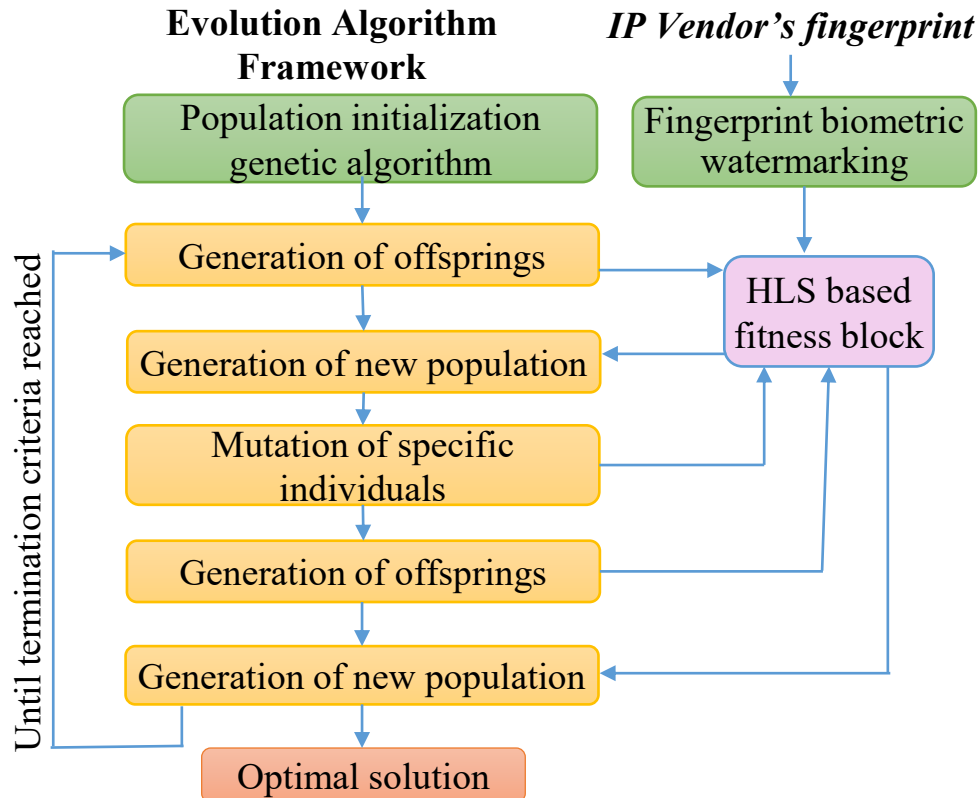


Fig. 3. DSE of secure optimal fingerprint watermark

Biometric fingerprint process :

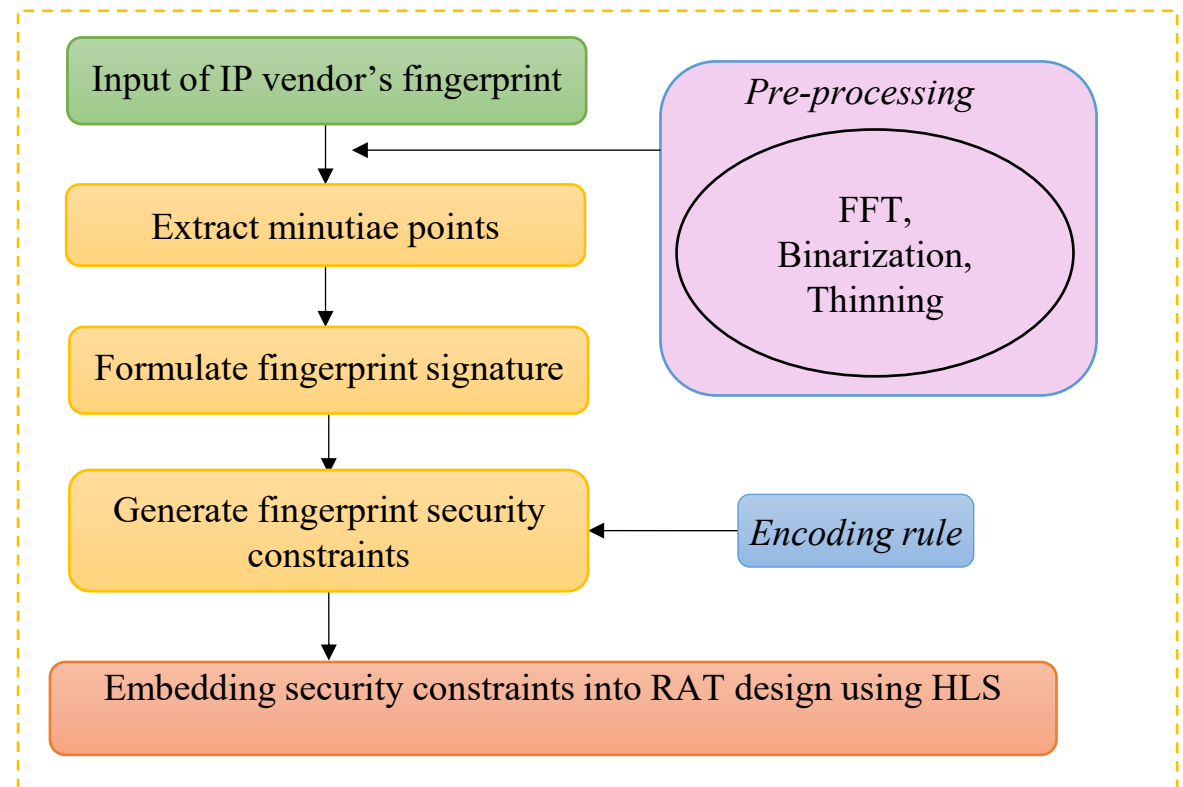


Fig. 4. Proposed biometric fingerprint based watermarking approach of secure IP core design

PROPOSED WORK CONTINUE...

Genetic algorithm: *Encoding process*

Input block					
$X_1 =$	$\alpha_{\max}$	$\beta_{\max}$	$\gamma_{\max}$	$\lambda_{\max}$	(Max. values)
$X_2 =$	$\alpha_{\min}$	$\beta_{\min}$	$\gamma_{\min}$	$\lambda_{\min}$	(Min. values)
$X_3 =$	α_{avg}	β_{avg}	γ_{avg}	λ_{avg}	(Avg. values)
$X_4 =$	α_{rnd}	β_{rnd}	γ_{rnd}	λ_{rnd}	(Random values) b/w Min and Max

Offsprings and their parents (X_i and X_j)	α	β	γ	λ
X_1 and X_2 : S_1	$\alpha_{\max}$	$\beta_{\max}$	$\gamma_{\min}$	$\lambda_{\min}$
X_1 and X_3 : S_2	$\alpha_{\max}$	$\beta_{\max}$	γ_{avg}	λ_{avg}
X_1 and X_4 : S_3	$\alpha_{\max}$	$\beta_{\max}$	γ_{rnd}	λ_{rnd}
	.	.	.	.
X_4 and X_1 : S_{10}	α_{rnd}	β_{rnd}	$\gamma_{\max}$	$\lambda_{\max}$
X_4 and X_2 : S_{11}	α_{rnd}	β_{rnd}	$\gamma_{\min}$	$\lambda_{\min}$
X_4 and X_3 : S_{12}	α_{rnd}	β_{rnd}	γ_{avg}	λ_{avg}

Fig. 5 Representation of gene encoding and offsprings (S_n) generation

PROPOSED MUTATION

- The following rules govern the mutation process:
 - For even count genetic individuals in the population, all elements (α , β , γ) are left-shifted, excluding λ ;
 - For odd count genetic individuals, random values within the specified range for each constraint are generated.
- The mutation algorithm is shown in Fig. 6.

PROPOSED MUTATION ALGORITHM

```
Input – Genetic individual
Output – Mutated genetic individual
For i=1 to n // where n=Max. population size in evolutionary algorithm
{
    If (i%2==0) // Left Rotation
    {
        for p=1 to D // D= No. gene dimension in  $X_n$ 
        {
            Temp =  $R_p$ 
             $R_p = R_{p+1}$ 
             $R_{p+1} = \text{temp}$ 
            p++
        }
    }
    if(i%2==1)
    {
        for p=1 to D
        {
             $R_p = R_p \pm Y$  // Y is a random number between [1,3]
            p++
        }
    }
    i++;
}
```

Fig. 6. Mutation algorithm

DEMONSTRATION

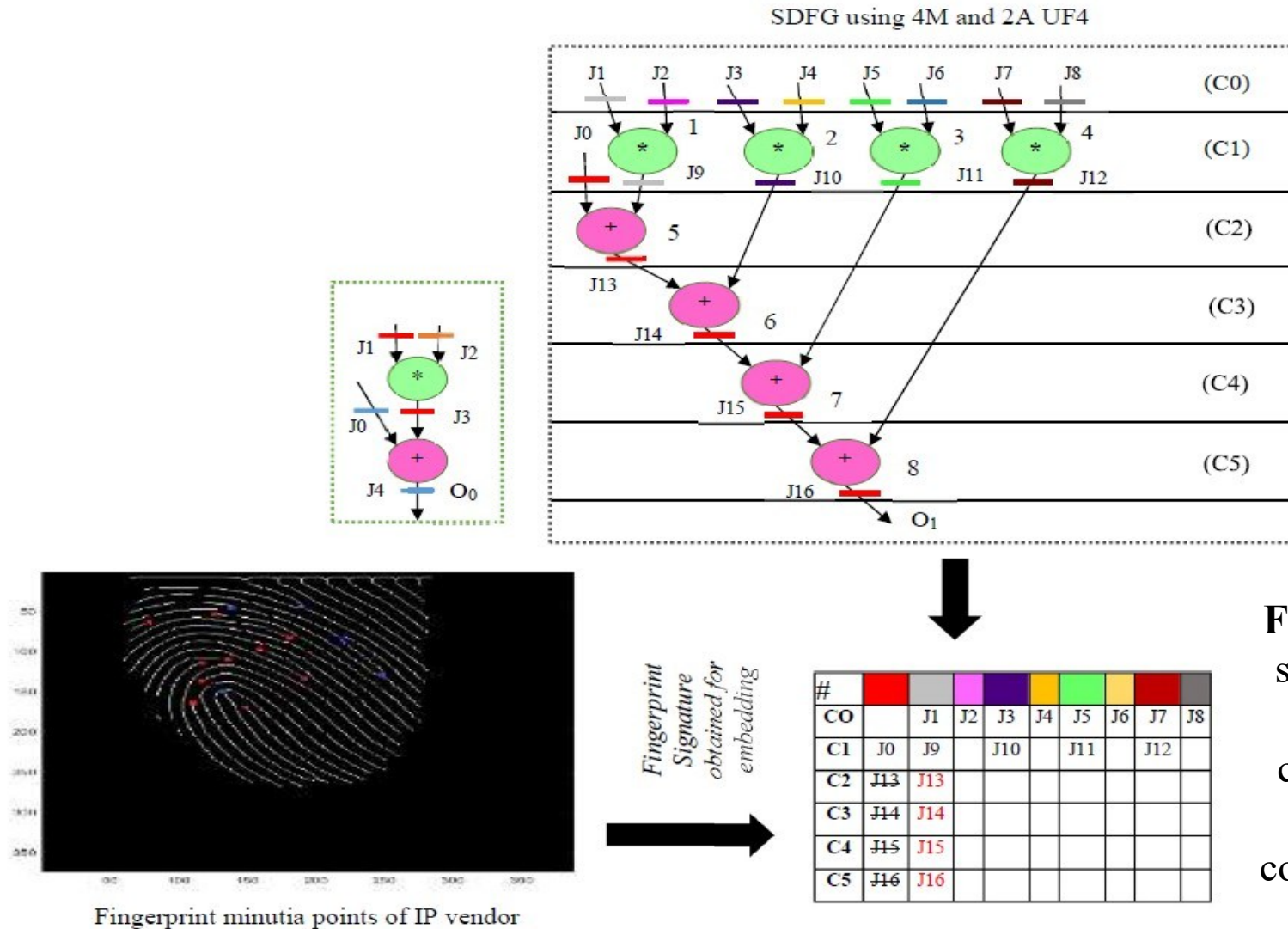


Fig.7. Demonstration of fingerprint signature embedding on 8 tap FIR filter RAT containing 9 colors/registers (obtained from its respective scheduling consuming 6 control steps C0 – C5)

RESULT AND ANALYSIS

Evaluation parameters :

➤ Probability of Coincidence :

$$P_c = \left(1 - \frac{1}{c}\right)^\lambda$$

Where 'c' represents the count of registers utilized before embedding the watermark security constraints, and ' λ ' signifies the number of encoded digits used in watermark security constraints.

➤ Design Cost :

$$CF = W1 * \frac{(Ar - Ac)}{Amax} + W2 * \frac{(L - Lc)}{Lmax}$$

Where, W1 and W2 = 0.5, Ar= area of each fingerprint watermarked design solution, Ac=area constraint value specified by the IP vendor, L= schedule latency of the fingerprint watermarked design solution, Lc= latency constraints value specified by the IP vendor.

RESULT AND ANALYSIS

Generations	Pre-mutation		Post-Mutation	
	Genetic individuals	Cost	Genetic individuals	Cost
G2	[4,8,1,140], [4,8,2,156]	0.2604 0.114	[4,1,1,140], [4,2,5,156]	-0.0898 -0.1861
G4	[4,8,1,140], [4,8,5,165]	0.2604 0.1651	[2,6,8,140], [2,2,6,165]	-0.059 -0.1418
G6	[4,8,1,140], [3,8,8,140]	0.2604 0.2476	[4,2,8,140], [4,1,3,140]	-0.1662 -0.0675
G8	[3,8,1,140], [3,8,8,140]	0.2476 0.1532	[3,3,3,140], [2,3,2,140]	-0.1255 -0.167
G10	[4,8,1,160], [3,8,1,140]	0.2604 0.2476	[4,1,4,160], [2,8,3,140]	-0.0898 -0.1171
G12	[4,8,1,160], [3,8,1,140]	0.2604 0.2476	[1,4,6,160], [4,7,8,140]	-0.0864 -0.0177
G14	[3,8,1,160], [3,8,1,140]	0.2476 0.2476	[4,1,3,160], [1,7,6,140]	-0.0675 -0.0293

TABLE 1. Pre and post mutation cost of the proposed approach for 8 tap filter

RESULT AND ANALYSIS

TABLE 2: Pc comparison wrt. IP vendor's watermark

Applications	Pre-embedding registers	Proposed Approach	DNA based hardware watermarking [5]
8 tap FIR filter	17	2.06E-4	4.26E-4
100 tap FIR filter	201	4.87E-1	5.28E-1

TABLE 3: Exploration time analysis of the proposed approach

Applications	For X=4	For X=6	For X=8
8 tap FIR filter	1.40 sec	2.18 sec	0.84 sec
100 tap FIR filter	7.97 sec	22.34 sec	26.78 sec

REFERENCE

- [1] A. Aldahmani, B. Ouni, T. Lestable and M. Debbah, "Cyber-Security of Embedded IoTs in Smart Homes: Challenges, Requirements, Countermeasures, and Trends," IEEE Open Journal of Vehicular Technology, vol. 4, pp. 281-292, 2023.
- [2] R. Heartfield et al., "A taxonomy of cyber-physical threats and impact in the smart home", Elsevier, Comput. Secur., vol. 78, pp. 398-428, 2018.
- [3] A. Sengupta, M. Rathor "Securing Hardware Accelerators for CE Systems using Biometric Fingerprinting", IEEE Transactions on Very Large Scale Integration Systems (TVLSI) , Vol: 28, Issue: 9, Sep 2020, pp. 1979-1992
- [4] A. Sengupta, S. Bhadauria, "Exploring Low Cost Optimal Watermark for Reusable IP Cores during High Level Synthesis", IEEE Access, Volume:4, Issue: 99, pp. 2198 - 2215, May 2016
- [5] A. Sengupta, R. Chaurasia "Securing IP Cores for DSP applications using Structural Obfuscation and Chromosomal DNA Impression", IEEE Access, Volume: 10, May 2022, pp. 50903 - 50913
- [6] F. Koushanfar, I. Hong, and M. Potkonjak, "Behavioral synthesis techniques for intellectual property protection" ACM Trans. Design Autom. Electron. Syst., vol.10(3), pp. 523–545, Jul. 2005.

Thank You!